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Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol

Tools to enhance public participation and confidence in the development of the Howard East aquifer water plan, Northern Territory

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ARTICLE INFO

Article history:

Available online 24 February 2012

Keywords:

Stakeholder analysis

Visualisation tools

Collaborative planning

SUMMARY

For the first time in the Northern Territory's water management history it is facing the challenge of managing an over-allocated groundwater resource. Rapid rural residential and agricultural development in the Howard River catchment of Darwin's hinterland has increased competition for groundwater from the bore-field that supplements the capital's metropolitan water supply. This has generated tensions between different water users and precipitated a water allocation plan for the Howard East aquifer. Initial context analysis indicated a widespread lack of public understanding of groundwater systems and processes, leading to misconceptions about the origin of local groundwater resources, groundwater–surface water interactions, extraction rates and impacts. In addition there is a legacy of mistrust by some peri-urban community members of government-driven planning processes to manage groundwater resources. The main objective of this study was therefore to trial two planning tools suited to this context over a 15 month period: (i) an extended stakeholder analysis and (ii) the participatory development of a 3D visualisation model, via a process described as the participatory Groundwater Visualisation Tool (GVT). The tools assisted the water planning agency to better understand stakeholder needs and interests, contributed to popular scientific understandings of hydro-geological conditions and processes, as well as captured local knowledge and values in preparation for an open and effective planning process.

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1. Introduction

Water management institutions are evolving in north Australia in the face of national-scale legal and economic reforms and in the context of variable hydrological systems (Straton et al., 2010; Tan, 2008). Several drivers are influencing the course of water management in Australia's tropics including increasing awareness of the full range of social, economic, environmental and cultural values for water, institutional changes to water allocation systems, intensifying land use and concomitant growing demand for water for irrigated agriculture and increasing uncertainty about water supplies in the face of climate change (Gehrke, 2005; Jackson et al., 2008). As demand for water grows, north Australian jurisdictions are progressing slowly towards compliance with the National Water Initiative (NWI) directives, yet water planning has been undertaken in only a relatively small number of catchments (Jackson and Altman, 2009; Tan, 2008). In the Northern Territory three water allocation plans have been completed.

2. Planning context

The Northern Territory's Howard River catchment covers approximately 1500 km² immediately adjacent to the capital city of Darwin (Fig. 1). The catchment has a mixture of land tenures, including leasehold for pastoral activities, mineral leases and permits for extraction and exploration, and freehold title. Land use within the catchment provides for peri-urban townships, Aboriginal and recreation sites, and horticultural and pastoral production. Water is sourced from private bore-fields and two bore-fields operated by the water utility provider (PowerWater Corporation). Current estimates of the population are 12,000 inhabitants (Australian Bureau of Statistics, 2007) with the population expected to at least double by 2021.

Approximately 90% of the region's monsoonal rainfall occurs during November to March, when lagoons, rivers, and stream channels are re-charged, as are the subsurface aquifers. Water levels in the aquifers, notably the shallow ones, gradually decline during the dry season and, with the exception of deep pools, the majority of waterways dry out completely (Padovan, 2003). As the supply of surface water decreases, the neighbouring peri-urban communities and the region's ecosystems increasingly rely on the groundwater source; the aquifer supplies a substantial amount of the Northern Territory's horticultural and vegetable industries

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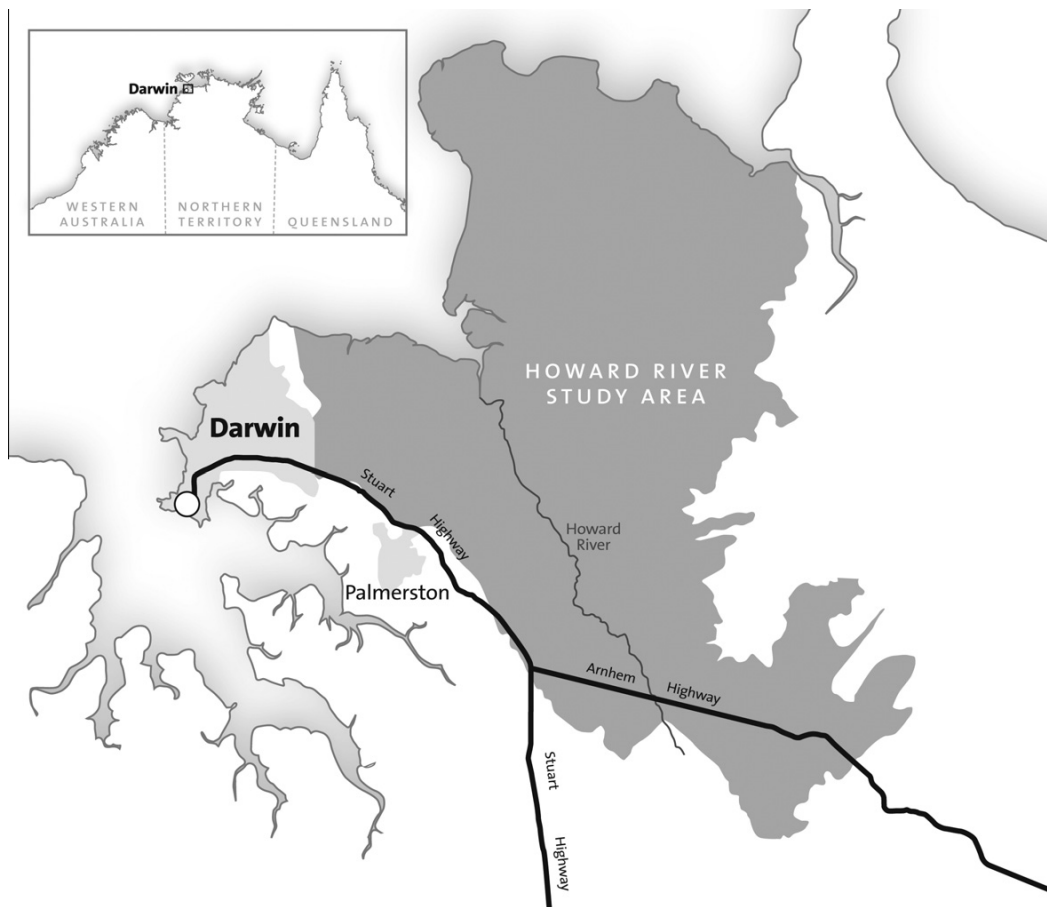


Fig. 1. The Howard catchment is defined to encompass the main subsurface dolomite aquifer and stretches east from the outskirts of Darwin's suburbs at Knuckey Lagoons to the Adelaide River floodplain, and north from the Arnhem Highway to the coast.

(Australian Bureau of Statistics, 2008) and 15% of Darwin's potable water (Northern Territory Government, 2008). Both groundwater and surface water flow provide a focus for public use of the ecological and cultural assets of the area; these flows sustain the many highly valued waterways, wetlands, rainforests, flora, and fauna utilised and appreciated by residents and visitors. Water therefore underpins the social and cultural values associated with the area, including activities such as: recreational fishing and hunting, cultural heritage, customary hunting and collecting, education and leisure (Woodward et al., 2008).

Population growth in and around Darwin has increased the number of private bores drilled in the area and this is expected to continue (Northern Territory Government, 2002), intensifying competition for groundwater between all water users, including the environment¹. According to Straton et al. (2010), tensions between urban and rural user groups are manifest:

Evidence of tension over water use is also seen in concerns over water flows in popular springs and the impacts of lower water flows and pollutants on groundwater dependent ecosystems (Straton et al., 2010, p. 145).

¹ Nolan et al. (2009) report that over 3300 productive bores were estimated to be in use within the Howard East area according to bore driller's records. Of these, 600 belong to commercial horticultural licence holders who irrigate treed crops, cucumbers and melons and vegetables (ABS, 2008). The remaining 2700 bores are largely unmetered and used for 'stock and domestic' purposes to support residential dwellings.

2.1. Status of planning

The Howard East aquifer has no previous history of water planning and for the reasons previously outlined in Section 1, the Northern Territory Government has now prioritised the development of water allocation plans for the rural regions of Darwin. Hydrological data and other information on water use in the Howard River region are being gathered to inform a water allocation plan. The region is part of the Darwin Water Control District and any plans developed within that district will be prepared under the Northern Territory's *Water Act 1992* (NT) which contains provision for community consultation and participation in plan development.

Preparing for increased pressure on water resources will require the collection and compilation of more detailed ecological, hydrological and socio-economic data across a wider geographic range, to support assessment of environmental flow requirements and to better understand surface–groundwater interactions. Plans will be built on information regarding:

- the aquatic environment, social and cultural values;
- how much water is required for different uses and when; and
- the potential effects of increased groundwater use and appropriate management actions.

In partnership with the Water Resource Management Branch of the Department of Natural Resources, Environment, the Arts and

Sport (NRETAS), the Howard East water allocation planning process was selected as one case study in the Collaborative Water Planning Project, a study pre-dating the Water Planning Tools project described fully in this special issue (see Tan et al., 2011, this issue).

An initial expectation of the collaboration was for the project to work closely with members of the Howard East Water Advisory Committee and planning staff to support their efforts to engage the broader community in water planning during 2009. A number of delays, however, affected the formation of a water advisory group, which was not constituted until late 2010 (<http://www.nt.gov.au/nreta/water/hewac/index.html>). The case study reported here trialled two planning tools: an extended stakeholder analysis and a participatory process referred to as the Groundwater Visualisation Tool (GVT), in full consultation with the rural community of the greater Darwin region.

3. Planning problem

The rights of stakeholders to participate in water use decisions are now widely recognised and often enshrined in policy frameworks, such as the Australian National Water Initiative (COAG, 2004) and the European Union Water Framework Directive (Commission of the European Communities, 2002), which urges states to involve interested parties as early as possible in planning processes (Straton et al., 2010). According to Straton et al. (2010), there is demand for tools that can (i) effectively include the range of disciplinary, stakeholder and institutional components that impact on resource planning (Wilson and Howarth, 2002), including ecological and social complexity (Wittmer et al., 2006); (ii) enable transparency (Lafferty and Meadowcroft, 1996; O'Neill, 2001) and participation in the sense of Renn et al. (1995); (iii) create decision-making institutions that encourage system learning and adaptability (Crosby et al., 1986; Meppem, 2000; Pearson et al., 2010); and (iv) enable social equity and procedural fairness (Habermas, 1984; Rawls, 1971; Sen, 1995). The tools trialled contributed to the goals encompassed by these stated objectives from the literature, particularly by enhancing transparency and improving understanding of social complexity.

In the Northern Territory, and indeed in much of Northern Australia, water planning has been undertaken in few catchments. In this study we built on research which found that, agency, community, and industry stakeholders in water planning recognised the importance of technical information and scientific research to water planning; but identified the absence of locally-held, community expertise as a knowledge gap (Tan et al., 2011, this issue). Therefore methods we adopted were especially targeted to support the development of skills for water planners in community engagement, identify information 'gaps' from the perspective of stakeholders and the community, and address the question of how best to incorporate non-technical information into the decision-making processes.

4. Tool trials

4.1. Extended stakeholder analysis

Considering the limited experience in water planning in the case study area, a stakeholder analysis was considered to be an important first step for 'setting the scene' for future community and stakeholder engagement (Nolan, 2009). In most water planning processes in Northern Australia, this is a quick, informal exercise. A more comprehensive stakeholder analysis (Brugha and Varvasovszky, 2000; Grimbale, 1998) would identify the actors that needed to be involved in local water planning, examine their water use interests and likely influence over the planning process, before

assessing their needs and capacity for meaningful participation. This extended process was initiated in the Howard East region.

A list of potential stakeholder groups in the Howard East region was created with government agency staff, local council members and Commonwealth Scientific and Industrial Research Organisation (CSIRO) researchers who had previous experience working in the case study area (Woodward et al., 2008; Straton et al., 2010). Other stakeholders were found through searches of local media to identify spokespeople and community 'experts' with an interest in water issues. Indigenous traditional owners, environmental, industry and recreational user groups were contacted directly. During an initial round of interviews, a contact list of potential stakeholders was shared with participants who were asked to add community stakeholders or spokespeople whom they thought should be contacted.

A series of 37 semi-structured interviews with key stakeholders from seven different community sectors was undertaken in late 2008. Open-ended questions were designed to (i) understand stakeholder concerns and attitudes toward water planning; (ii) identify trusted sources of information about the aquifer within the community; (iii) ascertain the level of interest and influence of different stakeholder groups to participate in future water planning processes; and (iv) determine their preferred modes of engagement. A short quantitative survey asked stakeholders to nominate the areas where they had enough information and/or areas where they would like more information. A range of secondary data resources were also drawn upon to build understanding of stakeholder and community hydrological knowledge and perspectives relevant to water management.

4.2. Groundwater Visualisation Tool (GVT)

After testing key findings from the stakeholder analysis in further community and stakeholder meetings, including with water planners; two specific barriers to collaboration were identified, and these informed the development of a second planning tool. The barriers were (i) a wide spread lack of understanding of groundwater systems; and (ii) a lack of trust of the science underpinning decision making and government driven management of the Howard East aquifer. The following quotes provide an insight into the problems identified during stakeholder interviews:

"The understanding of the resource is extremely poor. There's a lot of money going into community consultation and regulatory services but the things that underpin that ... the research on how much water is available for use and hydrology, are going down. I mean what my water use is, and what the value of the resource is – we are very poor in those areas", and

"I think one of the big issues that the authorities have to face is the fact that no one believes any of the recharge rates or the possible quantities of water available for use. Because they don't believe the science is there" (Field interviews 2009).

At the inception of the project, community members were wary of being associated with planning processes that were government driven. Many respondents believed that the Northern Territory Government intended to use information obtained from monitoring of bores to charge for the use of water. Community members were also looking for assurances that their participation and input into planning would be meaningful and influential.

Identifying these barriers led to the proposal for public involvement in a participatory process to assist the community to 'visualise' the groundwater systems, changes in groundwater levels over time, and to answer questions relating to increased water use in a variable environment by drawing on knowledge not readily accessible (Gartin et al., 2010). The complex hydrogeology and highly

variable yields of the aquifer was difficult for community members to understand. It was apparent to the research team and NRETAS that minimal agency-based information about the aquifer was trusted by the broader community, leading to common misconceptions about the origins and current state of groundwater resources. Further, the long-held view in this region is that groundwater is abundant as the large monsoonal rainfall replenished the regional aquifers on an annual basis, affecting community perceptions of the need for water planning. Some stakeholders for example, had difficulty in understanding the numerical-type hydrological models that showed shallow bores in danger of running dry during years of low rainfall. In contrast, water agencies were confident that the system was over allocated and not sufficiently replenished over time.

In order to create a 3D visualisation of the groundwater systems of the Howard East area, the Water Planning Tools project researcher collaborated with a team consisting of hydrogeologists and visualisation modellers at the Queensland University of Technology (QUT) and with groundwater personnel from NRETAS and the PowerWater Corporation. The agreed goal of the collaboration was to develop an integrated visualisation model using specialist software that could be installed and operated on household computers. This software, Groundwater Visualisation System (GVS²) (Hawke et al., 2009), can display the groundwater system framework and function and answer basic questions such as the origin of groundwater resources, the connectivity between ground and surface water, how rainfall and abstraction impact on aquifer recharge and the impact of increased demand for water.

While the visualisation produced did not have a predictive capability, it was expected to assist users to become more comfortable with “modelling” and appreciate the complexity of the science underlying the groundwater models produced by the water agency. This process aimed to build recognition among community members of the need for comprehensive water resource data collection and groundwater monitoring and trust in the agency's capacity to manage the resource, particularly its knowledge of the hydro-geological system. The visualisation tool approach was to use this software output to enable users to ‘see’ the aquifer's spatial configuration, show cross-sections of the aquifer at specific points of interest and view past water level measurements at different points in time across seasons and over 24 h periods (see Fig. 2) This capability would allow the user to observe a number of facets, including:

- the geographical structure and boundaries of the aquifer in easy-to-understand cross sections;
- measured time series of local and regional levels of extraction and water levels in observation bores over time and season;
- the relationship between rainfall and aquifer recharge over time and season, displayed using animation software;
- the growing number of productive bores in the area; and
- the relationship between bore depth and yield (Hawke et al., 2009).

Two steps were undertaken to build credibility and confidence in the Groundwater Visualisation Tool process. First, the visual model was built by independent hydro-geologists at Queensland University of Technology; who also participated in the public meetings and, second, a participatory approach was adopted to improve the community's trust in its veracity thus increasing the likelihood of adoption. Researchers developed a community engagement strategy that maximised the opportunities for local in-

put of hydrological knowledge into the process and the construction of the visualisation system. Information was obtained through a number of means, including a groundwater bore inventory (made available via the web), semi-structured interviews with local experts and participatory mapping exercises using high resolution topographic maps undertaken with local bore drillers and ex-agency staff. Rainfall data was also collected from residents of the area.³ The Queensland University of Technology personnel also spent time with NRETAS staff to obtain data and update their knowledge.

Key stakeholder groups, identified in the extended stakeholder analysis, were also invited to ‘road test’ the visual model and before its release were asked to comment on its accuracy and utility as a community education tool. The researcher (Nolan) had community development skills which were applied to the three different areas of engagement described in the following Sections 4.2.1–4.2.3.

4.2.1. Activities to engage government agency staff

Structured workshops, meetings and field trips proved useful in building a strong working relationship between project researchers, Queensland University of Technology modellers and hydrogeologists and staff from NRETAS and the PowerWater Corporation. This relationship was essential to enable the timely exchange of agency-held data sets at critical points of the Groundwater Visualisation Tool's development. Activities included a series of meetings and field visits with the Water Resource Management Branch hydrologists and modellers and a half-day workshop involving NRETAS staff and Queensland University of Technology modellers and hydro-geologist to gain feedback on model's utility and functionality.

4.2.2. Activities to engage local stakeholder groups

The researcher contacted those groups that the extended stakeholder analysis (see Nolan, 2009) identified as ones interested in participating in water planning and obtaining further information. Engagement activities included a series of meetings with local hydrologists, hydro-geologists, university researchers, modellers, local council members and water users. These activities were critical to developing a sense of ‘ownership’ and trust in the Groundwater Visualisation Tool process from its inception. In addition, the activities generated anecdotal information relating to water use, observed changes, whether there was seepage or not, the number of bores and when they had been drilled. Water users were interviewed with the aid of high-resolution topographical maps and the results were discussed at a workshop to gain feedback on the visualisation's utility and functionality. Two project newsletters were distributed and a dedicated project website established (<http://www.track.gov.au/project/howard-east>).

Stakeholders groups were encouraged to update their members by distributing project news or newsletters within their own networks and invite group members to community forums. An analysis of these activities indicated that each significantly improved the acceptance of the Groundwater Visualisation Tool among the stakeholder affiliated ‘community’, based upon feedback given in public meetings and emails, and the number of stories or discussion appearing in stakeholder newsletters and websites. As put by one stakeholder:

“The emails and newsletter updates were fantastic, a quick and easy way to see how the project was progressing” (Nolan et al., 2009, p. 16).

² GVS refers to the Groundwater Visualisation System software that is developed by QUT; GVT refers to the approach used by the Water Planning Tools team to use this as an interactive tool with the community.

³ Nolan et al. (2009) describe fully the knowledge collected from the community and its contribution to the model.

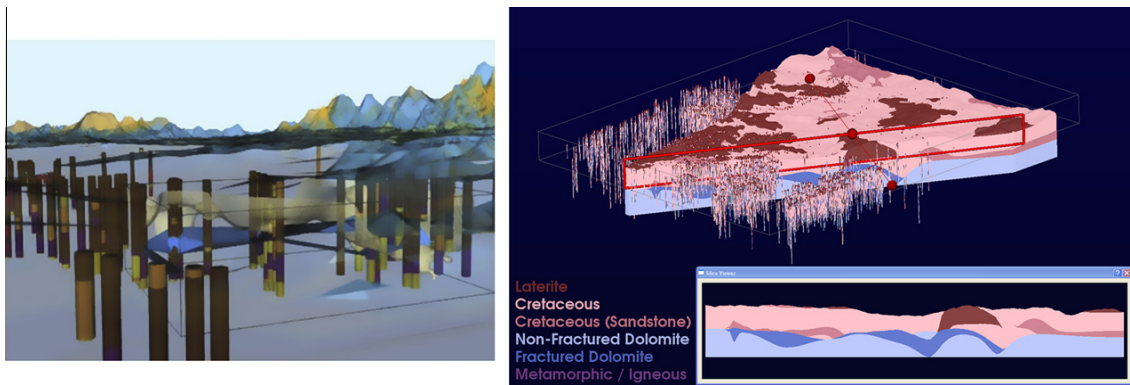


Fig. 2. Examples of images produced using GVS software.

When complete, copies of the visualisation tool were distributed to stakeholder groups with instructions outlining how it could be installed on household computers. Representatives were invited to a 'train the trainer' workshop to learn how to use the visualisation and demonstrate its utility to others.

4.2.3. Activities to engage the broader community

Given the contentious nature of water planning in the case study area, both researchers and agency staff thought it necessary to develop a strategy to engage members of the wider community whom may not be affiliated with an organised stakeholder group. Following development of the engagement strategy the project's activities were promoted at community forums, in posters and mail-outs, advertised in local and national radio programs, and project information stalls at local events were hosted.

Two community forums were held at the beginning and end of the Groundwater Visualisation Tool development. Each was considered to be well-attended by a representative cross section of community members (35 and 40 attendees respectively), generating a 'lively' discussion that provided researchers with an excellent forum for testing assumptions and gaining insight into local opinions. In the week leading up to each forum, invitations were extended via stakeholder websites, a large-scale mail-out (1200 printed invitations) and a media campaign featured on several local and national radio programs. Feedback from forum participants determined that promotion via radio was by far the most effective mechanism for community engagement. This finding was made further apparent when viewing the number of visits to the project website in the 2 days following interviews with researchers and local community members about the project aired by the public broadcaster (over 800 visitors).

Unlike the successes achieved using radio, efforts to engage the broader community in the Groundwater Visualisation Tool's development through informative posters, mailed invitations, project representation at prominent local events and a web hosted bore survey; yielded poor results. While the project website attracted over 2500 visitors in a 5 month period, few non-affiliated community 'stakeholders' downloaded the newsletter or reports on offer or responded to a call for voluntary information. Researchers considered that this lack of broader engagement can be associated with a common reluctance to be involved in planning issues that have yet to receive Ministerial endorsement. At the time of writing however, a timeframe for the planning process had been determined. As a result the trend of engagement observed during the research project may change.

These engagement activities were undertaken at different stages of the Groundwater Visualisation Tools's development to

optimise public participation and 'feed' community knowledge back into the model at specific times, as outlined in Fig. 3.

5. Results and outcomes

The Stakeholder Analysis was found to be of significant value to the water agency (Chris Wicks, pers. com.). It was first used to select representatives for the Water Advisory Committee formed in late 2010 (NRETAS relied heavily on the Analysis for a submission to the Northern Territory Cabinet). Excerpts from the document usefully contributed to a community engagement plan, and an agency issues paper to generate discussions about key issues of concern to be discussed by the Water Advisory Committee. Its recommendations have also led to the creation of two new roles within the Department of Natural Resources, Environment, The Arts and Sport, a full time Community Engagement Officer position, and a part time Stakeholder Manager position. It is still too early to evaluate the outcomes from the perspective of the Water Advisory Committee or other stakeholders.

The process for developing the Groundwater Visualisation Tool and its outcomes were evaluated by (i) a brief questionnaire administered at two points in time during the tool's development process and (ii) a focus group led by the project team at the completion of the model's development process, to elicit discussion around key points and draw lessons for future GVTs (Nolan et al., 2009). Questionnaire respondents reported that the participatory elements of the model (surveys, mapping exercises, newsletters and meetings) had increased the level of trust in the science underlying the Groundwater Visualisation Tool and consequently in the hydrological models of NRETAS. Ninety percent of GVT evaluation survey respondents indicated that they 'trusted' the science behind the model, knowing that local experts had been consulted in its development. Giving members of the community regular updates on the models progress and a range of options to get involved significantly improved the uptake and understanding of the final GVT. Stakeholders also stated that they would be more likely to use the model as an educational tool as it took into account the questions that the community were concerned about in the local area. Quotations from focus group discussions held to evaluate the tools are provided here to illustrate stakeholder views:

"I think that the greatest value is the visualisation – a picture tells a thousand words. You can talk to people/owners who have little concept of the system if you can show them or run them through a tool like this... I mean it's a great tool for drillers, or any horticulturalist thinking of developing" (Nolan et al., 2009, p. 22).

"It's going to give a holistic view to people who should think my bore is going to have an impact into the flows into Howard Springs."

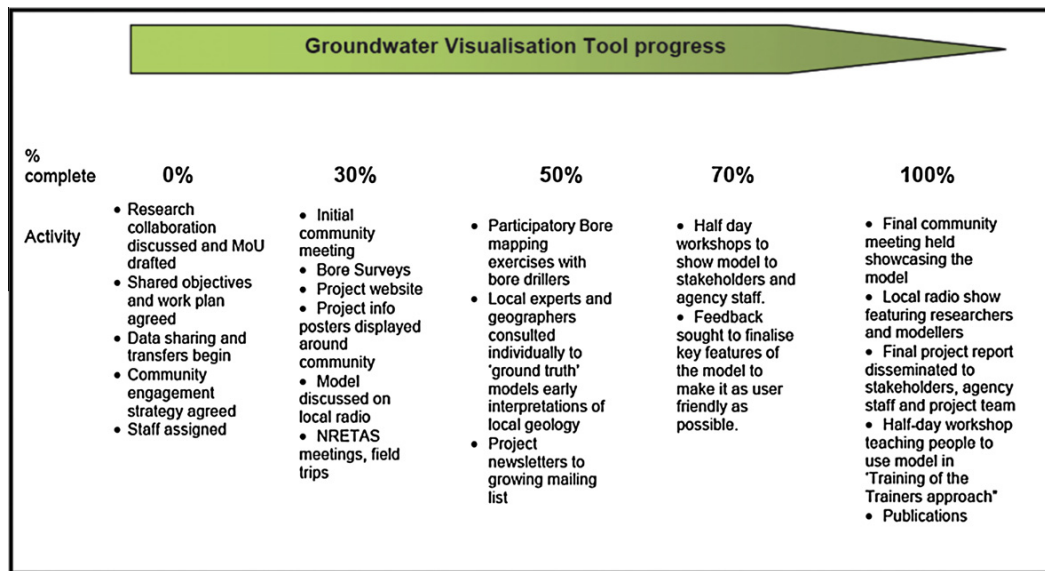


Fig. 3. Community engagement activities undertaken at different stages of development of the GVT approach.

That could be why we have bacteria issues there and they have closed it for swimming" (Nolan et al., 2009, p. 23).

"... the geology is complex and needs to be managed carefully" (Nolan et al., 2009, p. 24).

"It can show people all the data through a system that helps people relate a particular bore to its location and to the number of bores around it. People have to start thinking not in my own back yard – but rather 20 blocks around them – that's the sort of impact your activity can have" (Nolan et al., 2009, p. 23).

6. Discussion and conclusion

The participatory process that underpinned this research proved to be a source of strength as well as a challenge. On commencement of the project, researchers were expecting to work with a water advisory committee, in an active drafting phase for a water plan. Due to a delay to the start of the planning process beyond the control of NRETAS, we were compelled to redesign our research methods. On reflection, three observations may be drawn.

First, it was essential that there was adequate time to accurately map the expectations of stakeholders, understand their perceptions of the planning problem and note their preferences for engagement. The diversity of horticultural interests was revealed during the analysis, particularly the importance of immigrant horticulturalists from non-English speaking backgrounds. The final Water Advisory Committee includes three representatives from the horticultural sector (one from the NT Horticultural Association, a horticulturalist not aligned with the industry association and a representative of the Nursery and Garden Association) in an attempt to reflect the diversity of views. Stakeholder analysis is an area which planners often do not give sufficient attention to, and yet we think is essential to provide understanding of pre-existing conflict or issues which may jeopardise planning. In this case, the analysis had identified a widespread lack of understanding of groundwater systems and related pressures, and a lack of confidence in the science underpinning decision making. Subsequent planning effort could then focus on addressing this lack of knowledge and confidence. Taking a participatory research approach meant that we interacted with stakeholders and agencies

as co-researchers, addressing their expectations from the outset by developing shared objectives for the research and, by ensuring project information was made publically available and accessible. A dedicated project website, email contact, meetings, phone calls and project newsletters kept information flowing to interested parties over the course of 18 months.

Second, and equally as important as the first observation, the fusion of independent local and scientific 'experts' during the course of the trial was identified as a major achievement by the water agency (Ian Lancaster, pers. comm., Chris Wicks, pers. comm.). Drawing on local groundwater knowledge held within the community from the very outset of the planning tool approach, combined with community involvement in developing the visual models improved project legitimacy and public confidence. Discussions with several commercial groundwater drillers provided first-hand knowledge of groundwater occurrence in the area and the response of the systems to rainfall variations. Although there is a common-place perception that the layperson's knowledge has little to add to that already in the hands of a water agency, our work found that the independent expert, in this case the hydro-geologist from a partner university, was able to use local responses to shape his interpretation of data supplied by the agency. Rainfall data collected from residents was also drawn upon, but only to a minimal extent in this case. This type of anecdotal knowledge and community-derived data can be incorporated into the understanding of groundwater system processes and impacts of overuse. There are, however, limitations to the value of local knowledge if not collected in a consistent manner, as we found with bore survey data which was difficult to incorporate due to inconsistent driller descriptions.

Third, public confidence was further enhanced by independent input at meetings. Public meetings were chaired by an independent local Member of Parliament with strong facilitations skills. The support of a respected member of the local community was also useful because he was able to deflect more politically-charged questions that were beyond the responsibility of agency staff to address. The contribution of local bore drillers also instilled trust in the process.

The majority of participants considered the tools to be useful. The tools were found to be successful at analysing and addressing

community expectations that planning should be underpinned by credible information, and improved the overall capacity of the community to make informed decisions about groundwater management in the face of over-use of the resource. Mitigating against widespread and frequent adoption in water planning is the cost of development and implementation of the tools described here. The community engagement process can be expensive and time intensive, often requiring full time support throughout the project (or plan development). When an initial GVT is complete, there may be few resources left to sustain the community educational process or to update it with new information. In cases where conflict over water use is high and the knowledge base underpinning water allocations decisions is low or contested, we argue that the expense and effort of developing and implementing these participatory tools is warranted. Ideally, the tools could be delivered over a longer process by dedicated agency staff enabling them to establish long-standing links with the community and use the GVT to its full educational potential.

Acknowledgements

The Water Planning Tools Project would like to acknowledge the extensive contribution of time, effort, and expertise in building the GVS provided by Associate Professor Mal Cox and his team from the Faculty of Science and Technology, Queensland University of Technology, Brisbane, Australia. Acknowledgement is also due to the Queensland water agency (DERM) for co-funding the project in the Condamine, to DERM and NRETAs for provision of data, and to stakeholders in the Howard East for sharing their knowledge about groundwater. Wendy Proctor and Kim Alexander of CSIRO provided helpful comments on a draft of this paper as did an anonymous reviewer. Barbara McKaige of CSIRO greatly assisted the research team in its communication activities in Darwin.

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